Practice: equations and intersections

1. Solve $\cos(x) = -1/2$ for $0 <= x < 360$ degrees.

2. Solve $\log_{10}(4*x +968) = 3$. State the domain restriction.

3. Find the intersection of $y = -3*x -9$ and $y = 4*x +19$. Show your working.

Solutions

1. x = 120, 240 degrees. Use the unit circle; 360 degrees is excluded.

2. Domain: 4*x +968 > 0.

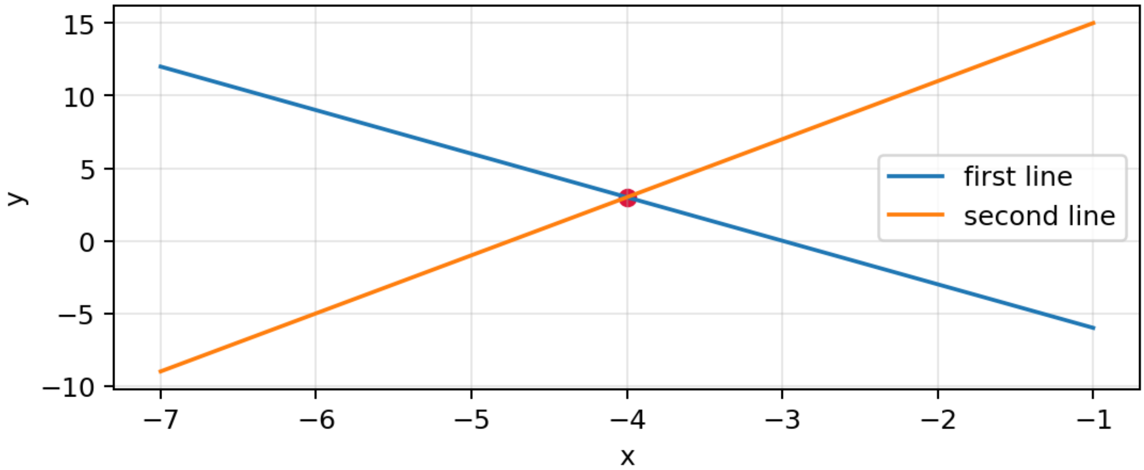
Rewrite: 4*x +968 = 10^3 = 1000.

x = 8. Check: the logarithm argument is 1000 > 0.

3. Set the expressions equal: (-7)*x = 28.

x = -4; substitute into either equation to get y = 3.

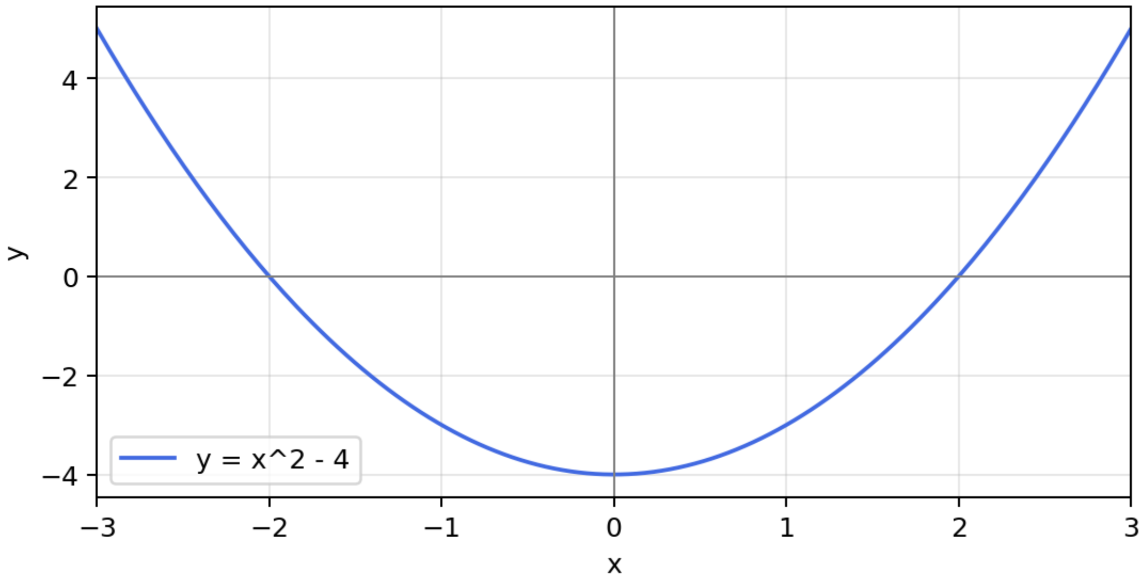
Intersection: (-4, 3).



Explore a quadratic

$f(x) = x^2 - 4, \qquad f\prime(x) = 2x$

4. Use the graph to identify the roots, minimum, and intervals of increase and decrease.



Solutions

4. Roots: -2 and 2. Minimum: (0, -4). Decreasing for $x < 0$; increasing for $x > 0$.